

SYSTEMATIC TRADING. MTH9897

Lecture 1

Quantitative Investment Framework, Pt.1



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US Investing Landscape.

- ❑ EMM (HFT) – short horizon
 - ❑ Tech-driven: fastest tick-to-trade strategy takes it all (80%)
 - ❑ Live or die by trading
 - ❑ No credible simulators of providing liquidity on public venues
 - ❑ Hence: provide liquidity to retail or take liquidity on exchanges/dark
 - ❑ Increasingly needing medium-frequency signals to manage inventory
 - ❑ Over-capacity, in search of new venues (crypto, prediction mkts). Mostly internal money
- ❑ StatArb (medium-to-long horizon)
 - ❑ Prop trading firms and quant HFs
 - ❑ Always at capacity, trading prowess is a key
 - ❑ Signals decaying, constantly in search of new (expensive) data
- ❑ Factor/Fundamental Investment (long horizon)
 - ❑ Bulk of retail and institutional funds (pensions, sovereign, endowments)
 - ❑ Overcrowded, but it may not be an issue, performance is relative to peer groups.
- ❑ Indexing (long horizon)
 - ❑ Increasingly popular, especially with retail (ETF proliferation)
 - ❑ Difficult (if not impossible) to beat long-term after-cost performance
 - ❑ Managing turnover and handling rebalances are more important
 - ❑ Tax-aware and levered variants offer attractive after-cost, after-tax returns.

Hedge Fund Industry: \$4.5T AUM

- ❑ Discretionary Equity L/S \$1.1T
- ❑ Multi-Strategy \$0.8T
- ❑ Credit \$0.6T
- ❑ Event-driven (M&A, special situations, activist) \$0.6T
- ❑ Discretionary macro \$0.5T
- ❑ Managed futures (Systematic) \$0.3T
- ❑ Quant Equity \$0.3T

Quant HFs ~0.7T, ~17% of HF industry assets

What is Alpha?

- ☐ Factors vs Alpha
- ☐ "Alpha factors" vs "Risk factor" vs "Pure Alpha" (e.g. idiosyncratic returns)
- ☐ What investors are paying for? Which investors?
- ☐ The role of crowding factors
- ☐ What if factor is known (from literature)?
- ☐ The role of (reliable) data. Factor performance decay.

BEFORE WE DELVE INTO QE: definitions

Time Series vs. Cross-Sectional Trading Strategies

☐ Time Series

- A. Intraday Statistical Arbitrage / HFT
- B. Agency Algos

Features:

- Each stock (or small group of stocks) is evaluated and traded independently of the rest of tradable universe
- ETFs/Futures are used primarily to hedge the exposures

Pros:

- Take advantage of asynchronous arrival of data

Cons:

- **Typically** doesn't evaluate an opportunity within full context of risk drivers. Hedging is basic
- Almost by design is focused on mean-reverting high-turnover strategies. (Exceptions are less popular technical momentum indicators – more expensive to trade)
- Intraday data is a small subset of available data. And is it highly commoditized.

☐ Cross-Sectional Regressions (medium-term statarb, canonical quants)

☐ Hybrid solutions (these are used as well)

Fama-MacBeth 2 stage regression

- ❑ Use **time series** data to regress each asset against the proposed risk factors to determine that asset's beta (β^j) for that risk factor.

$$R_t^j - R^f = \alpha + \beta^j (R_t^m - R_t^f) + \epsilon_t^j$$

- ❑ Use **cross-sectional data** to regress all asset returns for a fixed time period against the estimated betas (β^j s) to determine the risk premium b for each factor.

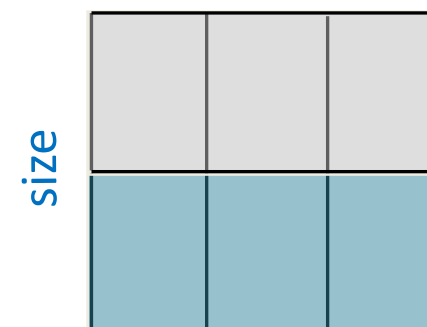
$$R_{next\ period}^j = a + b\widehat{\beta^j} + e^j$$

Fama-French 3-Factor Model

Form 2x3 portfolios

- A. Size factor (SMB)
 - Return of **small** minus **big**
- B. Book/Market factor (HML)
 - Return of **high** minus **low**

book/market



CAPM: for ... $R_t^j - R_t^f = \alpha^p + \beta^p (R_t^m - R_t^f)$

α^p are large and β^p do not vary much

FF3: for ... $R_t^p - R_t^f = \alpha^p + \beta^p (R_t^m - R_t^f) + \gamma^p SMB_t^p + \delta^p HML_t^p$

(for each portfolio p using time series data)

α^p are statistically indistinguishable from zero, while γ^p , δ^p coefficients are significant, higher R^2 . (Fama-French 3-factor model has been shown to explain ~90% of the returns of diversified portfolio, while CAPM explains only 70%). For 4 & 5 factor model (+momentum & quality)

<http://www.econ.yale.edu/~af227/pdf/Buffett's%20Alpha%20-%20Frazzini,%20Kabiller%20and%20Pedersen.pdf>

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2287202

- “Judgment Under Uncertainty: Heuristics and Biases” by D. Kahneman, A. Tversky
- “The Little Book That Still Beats the Market” by Joel Greenblatt
- “The Handbook of Equity Market Anomalies”, by Leonard Zacks (editor)
- “Quantitative Equity Portfolio Management”, by E. Qian, R. Hua, E. Sorensen
- “Quantitative Portfolio Management: The Art and Science of Statistical Arbitrage”, by Michael Isichenko (2021)

ANOMALIES: REAL VS NOT

- ❑ If you found a person among 1 billion people who threw 30 heads out of 30 coin tosses, what is the chance that next time his hit rate is $>50\%$?
- ❑ <http://www.wallstreethorizon.com/upload/IsThereNewsLatestJan2015.pdf>
 “It shows that when firms advance their earnings announcements at least four days prior to expectations, the earnings surprises in those quarters tend to be positive and the abnormal returns from two days after the earnings release date was announced through one day after earnings are actually announced are positive and significant. The converse is true for firms that delay their earnings announcement at least four days relative to prior expectations. “

So companies are delaying telling the bad news.... Does it seem like a common, recurring behavioral pattern?

❑ Risk mis-measurement:

- Factor is subsumed by known risk factors – “horse race” against others
- E.g. only shows ER (excess return – e.g. statistically significant risk premium) in univariate regression or CAPM, but not in Fama- French model (1993) – any decent economic paper tests for at least FF3,4,5.

❑ Statistical reliability

- Type I error: null hypothesis of zero abnormal returns is falsely rejected

❑ Data-snooping

- Finding accidental pattern in the data vs. accidentally finding a real pattern
- Real factors typically has economic rationale or theory, such that a series of tests can be designed to validate it (e.g. validate behavioral patterns, **not returns directly**).

WHAT IS A “REAL” FACTOR ANYWAY?... ARE ML FACTORS “REAL”?

IS ANOMALY EXPLOITABLE?

❑ Limits to Arbitrage

- Transaction Costs
- Short Sale Constraints
 - Always calculate ER for long- and short- portfolios
 - Check for hard to borrow names

❑ Arbitrageur presence

- Absence of Close Substitutes
- Unsalable opportunity (regulatory barriers, information leakage)

THEORIES OF FINANCIAL ANOMALIES

Standard finance is built on five foundation blocks:

- ❑ People are rational
- ❑ People construct portfolios as described by mean-variance portfolio theory (seeking highest return per unit of risk)
- ❑ People save and spend as described by standard life-cycle theory, where people find it easy to find and follow the right way to save and spend
- ❑ Expected returns of investments are accounted for by standard asset pricing theory, where differences in expected returns are determined only by differences in risk
- ❑ Markets are efficient, in the sense that prices equal values in them and in the sense that they are hard to beat

THEORIES OF FINANCIAL ANOMALIES

According to behavioral finance:

- ❑ People are normal, e.g. subject to cognitive and emotional biases
- ❑ People construct portfolios as described by behavioral portfolio theory, where people's portfolio wants extend beyond high expected returns and low risk, such as for social responsibility and social status (ESG...)
- ❑ People save and spend as described by behavioral life-cycle theory, where impediments, such as weak self-control, make it difficult to find and follow the right way to save and spend (MEME stocks...)
- ❑ Expected returns of investments are accounted for by behavioral asset pricing theory, where differences in expected returns are determined by more than differences in risk, such as by levels of social responsibility and social status
- ❑ Markets are not efficient in the sense that prices equal values in them, but they are efficient in the sense that they hard to beat (NVDA P/B = 49.32, GS P/B = 1.44 ... AMC's ???)

✓ Framing

People react to a particular choice in different ways depending on how it is presented; e.g. as a loss or as a gain. People tend to avoid risk when a positive frame is presented but seek risks when a negative frame is presented. (Penalty for late registration vs. discount for early registration. Sell winners too early, hold losers too late – so called disposition effect): **What is the price impact?**

✓ Hindsight

Tendency to repeat actions that brought good outcomes and avoid actions that brought bad ones (**momentum strategy?**). Breaks when there is no associations between past and future events or between actions and outcomes.

✓ Confirmation

Tendency to overlook or assign little weight to disconfirming facts and opinions. Classic: proverbial dog believes that his bark makes UPS truck go away. Test: not to bark next time and see if truck stays. Finance: I won = I'm skilled, I lost = bad luck. Tests on actual skills are rarely done. **What is the price impact?**

Why are we talking about retail investors and human decisions?

Who is losing money?

Cognitive Shortcuts and Errors, Cont'd

✓ Anchoring and Adjustment

Economic forecasts are adjusted from the previous quarter. 52 week high-low prices get undue attention (odds of stock downgrades by analysts when price is near 52-week high is 33% higher than the odds of downgrades at other times)

✓ Representativeness

“If a thing looks like a duck, swims like a duck, and quacks like a duck, then it must be a duck”. **Technical analysis** relies a lot on what might seem like a representative pattern from the past.

✓ Availability

Do dolphins really save drowning people or this belief is based on lucky survivors who they happen to push to the shore? Finance: survivorship bias. Amateur investors are frequent buyers of **attention grabbing stocks**.

✓ Overconfidence

(1) Overestimation – expecting higher portfolio returns than the average (2) Over-placement – thinking as being in top 10% of investors, while objectively barely making top half (3) Over-precision – underestimating volatility, e.g. expecting narrow outcome.

Emotional Shortcuts and Errors

✓ Hope and Fear

High past returns are associated not only with increased return expectations, but also decreased risk perceptions. (money won in a casino is put in a “separate” pocket).

✓ Happiness, Sadness, Disgust

Sadness among mutual fund managers was associated with subsequent lower returns. Disgust among public towards financial institutions or corporations was associated with selling their stocks at low prices.

✓ Anger

Linked to underestimation of the likelihood of losses. Leads to excessive risk-taking

✓ Regret and Pride

Linked to self-attribution of wins. Subsequent repurchases of previously sold stocks only if gains are realized (doesn't add to performance).

✓ Affect

Embrace stocks of admired companies, expecting high returns, distancing ourselves from stocks of spurned companies, expecting low risk. (TSLA/AAPL/AMZN ...?)

In addition to investor irrationality, consider rational **structural uncertainty**, e.g. investor uncertainly about economic environment:

- ✓ Having incomplete information
 - ☐ Short History
 - ☐ Small caps
 - ☐ Relaxed accounting standards

- ✓ New “disruptive” technologies
 - ☐ No frame of reference
 - ☐ Uncertainly about valuations parameters

...difficult to separate these categories. Overconfidence belong to both...

When was the last time we lived through the current macro-regime...?

Finfluencers (May 2023)

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4428232

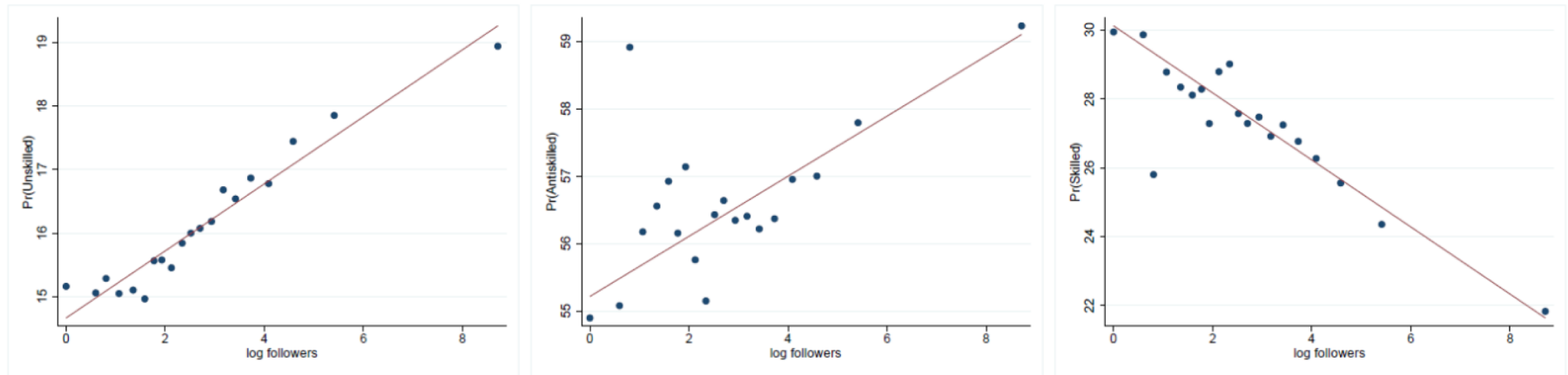


Figure 3: Binscatter Plots Skill versus of Number of Followers

The plots show binscatter plots of the probabilities of users being skilled, unskilled, and antiskilled, respectively, versus the natural logarithm of the number of followers.

Tweet-level data from a social media platform reveals low average accuracy and high dispersion in the quality of advice by financial influencers, or “finfluencers”: 28% of finfluencers are skilled, generating 2.6% monthly abnormal returns, 16% are unskilled, and 56% have negative skill (“antiskill”) generating -2.3% monthly abnormal returns. Consistent with homophily shaping finfluencers’ social networks, antiskilled finfluencers have more followers and more influence on retail trading than skilled finfluencers. The advice by antiskilled finfluencers creates overly optimistic beliefs most times and persistent swings in followers’ beliefs. Consequently, finfluencers cause excessive trading and inefficient prices such that a contrarian strategy yields 1.2% monthly out-of-sample performance

Quantitative Investment Process

- Alpha Model (forecasts excess return of stocks)
 - Easier to find random factors that represent non-compensated market risk (beta) than to find alpha factors representing incremental rewards
- Risk Model
- Portfolio Construction / Optimization
- Portfolio Implementation
 - Minimization of implementation shortfall under uncertainty
 - Designing trading process that is “aware” about underlying models and portfolio constraints
- Performance Attribution
 - How much is working and how much is random. Relate ex-post returns to ex-ante factor exposures.

Return Forecasting. Typical Factors

	Theme	Representative Metric	What It Measures
Alpha Factors	Deep Value	P / Book	Measures attractiveness under “mean reversion” in fundamentals
	Current Value	P / Earnings	Measures attractiveness of a stock on current earnings
	Capital Use	Dividend Yield	Measure of shareholder “friendliness”
	Quality	Accruals	Measures earnings quality, accounting
	Profitability	ROE	Measures “internal” rate of growth, pricing power
	Momentum	Price Momentum	Measures investor reaction, fund flows, sentiment feedback
Control Factors	Beta	Beta	Measures predicted beta estimated by risk model
	Size	Market Cap	Measures magnitude of “size bias”
	Risk	Residuals	Measures possible embedded risk that can not be controlled

More “esoteric” control factors: exposure to highly-shorted names, MEME/retail stocks
HF crowded names, macro factors (e.g. rates, elections, Covid, etc.)

Three-Factor Model: size, book-to-market and excess return of the market

“Multifactor Explanations of Asset Pricing Anomalies”, E. Fama, K. French (1996)

- ❑ Book-to-market, a.k.a. “deep” or liquidation value
- ❑ Typical Definition: $P / B = \text{Market Cap} / \text{Book Equity}$
- ❑ Market Cap = Price x Shares Outstanding
- ❑ Book Equity = Total Asset – Total Liability
- ❑ Strategy: “Buy low, Sell High”

Why are we still talking about FF model? Isn't it some ancient stuff from 1990th?

Deep Value: Should I long MS and short KKR?

Company	Market Cap	P/B	ROE
JPMORGAN CHASE & CO	\$ 178,992,250,000	0.9	11%
CITIGROUP INC	\$ 137,021,780,000	0.7	7%
BANK OF AMERICA CORP	\$ 126,188,950,000	0.5	2%
GOLDMAN SACHS GROUP INC	\$ 66,609,500,000	0.9	10%
BLACKROCK INC	\$ 43,504,790,000	1.7	10%
MORGAN STANLEY	\$ 40,348,800,000	0.7	0%
BANK OF NEW YORK MELLON CORP	\$ 31,361,080,000	0.9	8%
STATE STREET CORP	\$ 26,010,240,000	1.3	10%
BLACKSTONE GROUP LP/THE	\$ 22,959,130,000	4.3	4%
SCHWAB (CHARLES) CORP	\$ 20,946,190,000	2.2	9%
CME GROUP INC	\$ 19,684,590,000	0.9	4%
T ROWE PRICE GROUP INC	\$ 19,285,390,000	5.0	23%
AMERIPRISE FINANCIAL INC	\$ 14,686,610,000	1.6	12%
KKR & Co LP	\$ 13,551,990,000	6.9	29%
NORTHERN TRUST CORP	\$ 12,560,100,000	1.7	9%
TD AMERITRADE HOLDING CORP	\$ 10,586,770,000	2.4	14%

Yes, this is from GFC period.... Can we ask the same question between TSLA (P/B =10.9x ROE=19.9%) & LCID (2.5x, -66.4%)?

Deep Value: Should I long SU and short EPD?

Company	Market Cap	P / B	ROE
EXXON MOBIL CORP	\$ 391,815,310,000	2.3	24%
CHEVRON CORP	\$ 225,158,670,000	1.7	19%
CONOCOPHILLIPS	\$ 70,194,880,000	1.5	17%
OCCIDENTAL PETROLEUM CORP	\$ 64,137,230,000	1.6	12%
ENTERPRISE PRODUCTS PARTNERS	\$ 55,502,900,000	4.2	18%
SUNCOR ENERGY INC	\$ 41,666,910,000	1.1	11%
ANADARKO PETROLEUM CORP	\$ 40,650,740,000	2.0	5%
KINDER MORGAN INC	\$ 40,256,450,000	2.9	8%
Phillips 66	\$ 35,870,160,000	1.7	19%
KINDER MORGAN ENERGY PRTNRS	\$ 32,485,110,000	2.8	18%
EOG RESOURCES INC	\$ 30,844,760,000	2.4	4%

As of 9/17/24: EPD P/B=2.25, ROE=21.1%, SU P/B=1.59, ROE=14.0%

➤ What is in a firm's Value?

- DDM: a firm's value is equal to sum of all future discounted dividends

➤ DDM

$$Value = \sum_{k=1}^{\infty} \frac{D_k}{(1+r)^k}$$

➤ Stable Growth Company ("Gordon Growth Model"): assuming constant dividend growth rate g and cost of equity r

$$Value = \frac{D_1}{(r - g)}$$

➤ Where does growth come from:

- Improving efficiency: increasing ROE (episodic, trends to zero as a firm matures)
- Re-investment

- Growth is closely linked to dividend payout ratio and ROE

$$g = ROE_t \times (1 - \text{payout ratio}) + \frac{ROE_{t+1} - ROE_t}{ROE_t}$$

- As firm matures

$$g = ROE_t \times (1 - \text{payout ratio})$$

- So how much dividend a firm can pay depends on earnings growth and efficiency

$$\text{payout ratio} = 1 - \frac{g}{ROE}$$

$$\text{dividend} = \text{net income} \times \text{payout ratio} = \text{Book Equity} \times ROE \times \left(1 - \frac{g}{ROE}\right)$$

- Putting them all together with :

$$\text{Firm Value} = \frac{\text{dividend}}{(r - g)}$$

$$\text{Firm Value} = \text{Book Equity} \times \frac{ROE - g}{r - g}$$

- If market is fair

$$\frac{P}{B} \propto \frac{ROE - g}{r - g}$$

- Where r is “cost of equity”. From CAPM

$$r = r_{free} + \beta \times r_{equity}$$

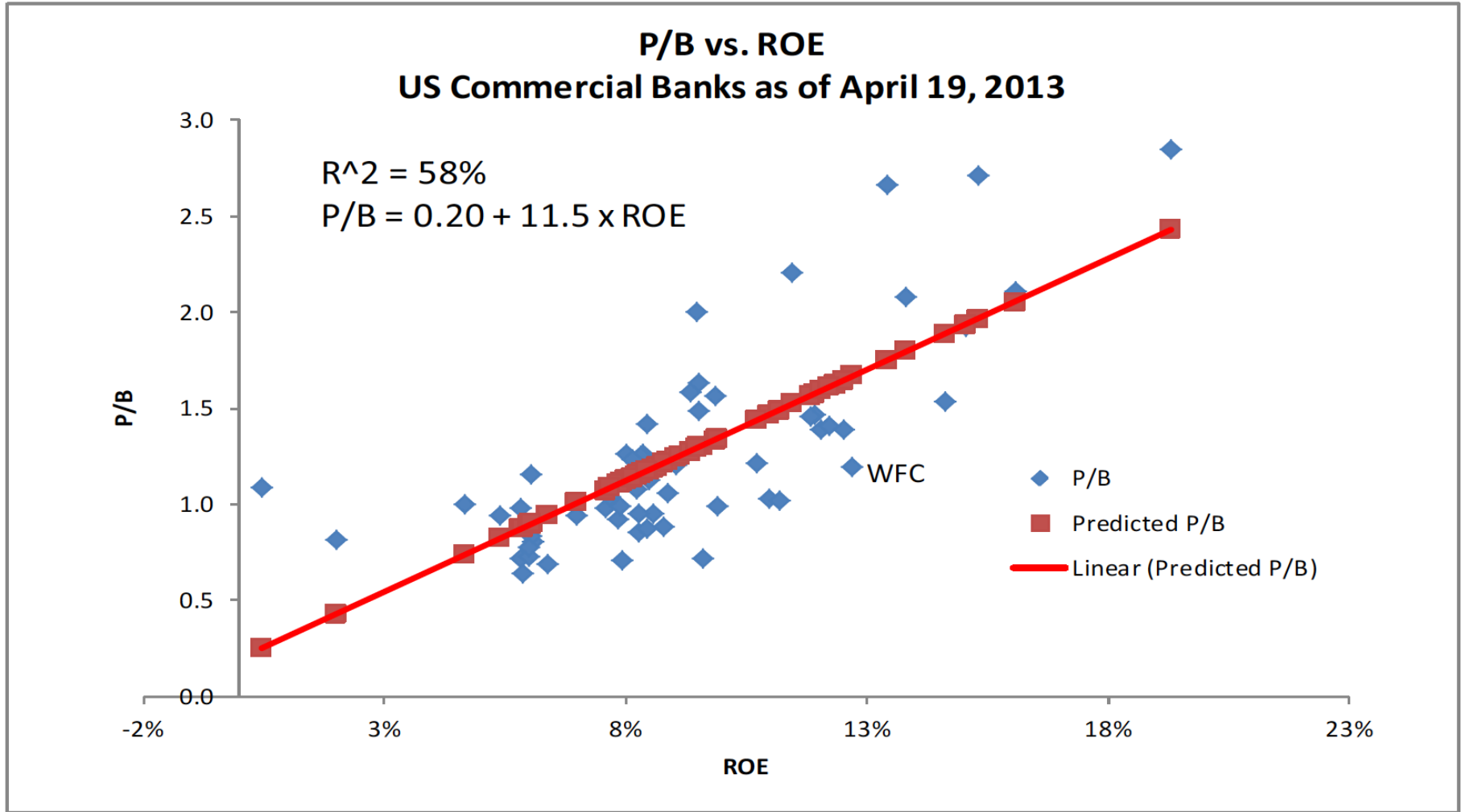
- This means that

- P/B is proportional to ROE
- P/B is inversely proportional to β

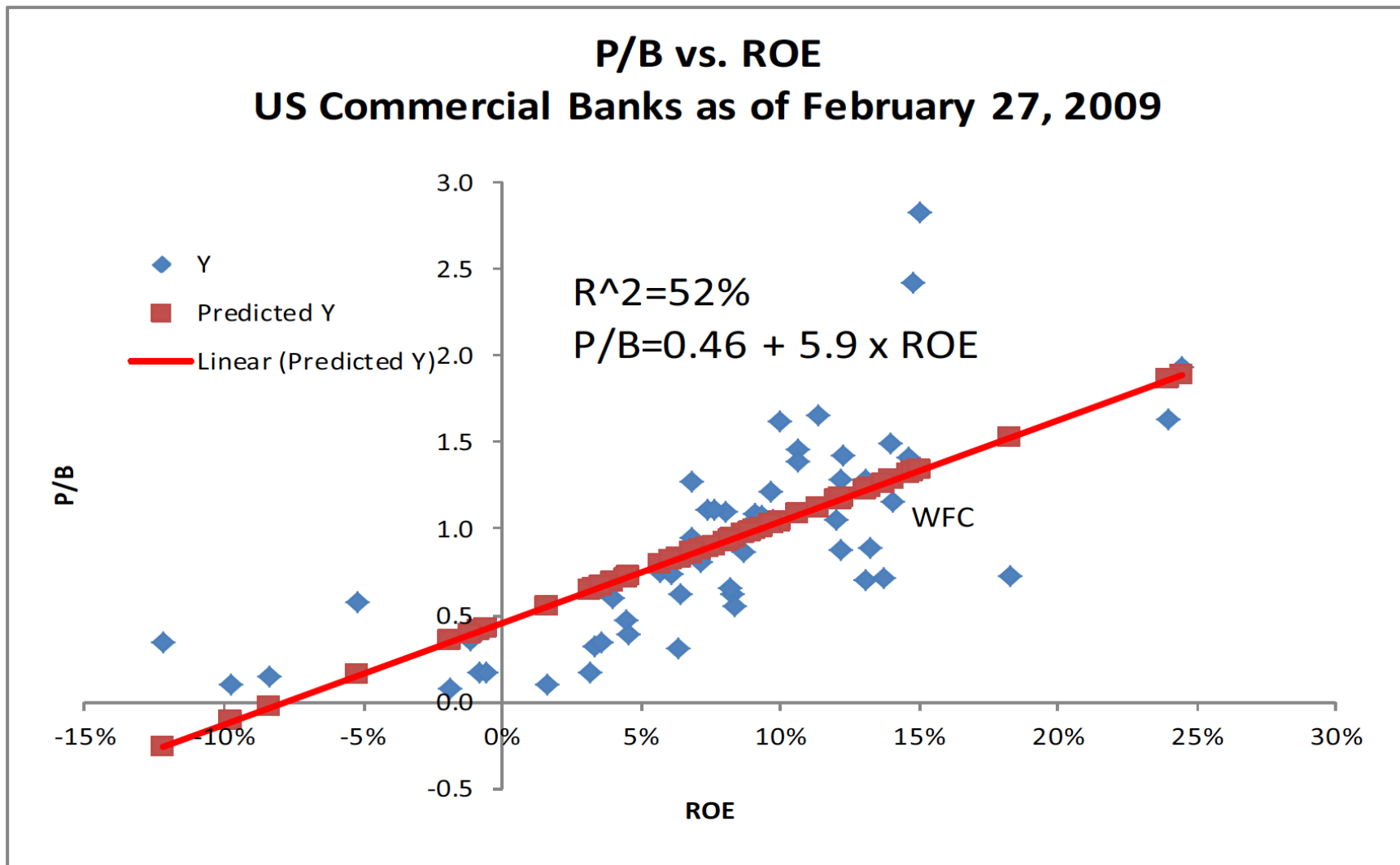
(1) Holding B , ROE , g constant, lower P implies higher expected returns r

(2) Holding P/B , g constant, higher ROE implies higher expected returns r

(3) Investments are either funded by earnings (reducing dividends) or funded externally (increase in liability & decrease in book value) implying lower expected returns



Case Study: US Commercial Banks, as of February 27, 2009



- Regress P/B against ROE (April 19, 2013)

$$\frac{P}{B} = 0.2 + 11.5 \times ROE$$

$$R^2 = 57\%$$

F – statistics: 81

- Adding *Impact of cost of equity*:

$$\frac{P}{B} = 0.8 + 10 \times ROE - 0.5 \times \beta$$

$$R^2 = 62\%$$

F – statistics: 47

Implications for deep value investors

- P/B is mainly an express opinion on the a firm's ROE
 - Investors pay up for high profitability, more importantly, sustainability of such profitability;
 - Investors pay fraction of book value when they believe the profitability of a company, or even an industry is permanently damaged;
- Other factors complicate the value proposition by P/B
 - Beta: higher beta → higher cost of equity → lower P/B, when all else being equal
 - Sector biases: more often than not, disbelief in sustainability of profitability isn't isolated to just one firm but the entire industry ("Investment Banks" a case in point after 2008)

Recent (2023) examples: (1) what happened to regional banks in spring? (Silicon Valley, Signature, First Republic, Hearland Tri-State, etc.)

(2) What about companies that keep transforming themselves? Netflix? Amazon?

A five-factor model directed at capturing the **size, value, profitability, and investment patterns** in average stock returns is rejected on the GRS test, but for applied purposes it provides an acceptable description of average returns. The model’s main problem is its failure to capture the low average returns on small stocks that invest a lot despite low profitability. The performance of the model is not sensitive to the specifics of the way its factors are defined, at least for the definitions considered here.

- ❑ Use of DDM to explain the role of B/M, profitability and investment to predict future returns
- ❑ B/P is a noisy proxy for expected returns, because P also responds to forecast of earnings and investments.
 - ✓ Novy-Marx (2012) “The Other Side of Value: The Gross Profitability Premium” <http://rnm.simon.rochester.edu/research/OSoV.pdf>
 - ✓ Aharoni, Grundy, Zeng (2013) https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1800603
- ❑ The results reveal that the highest returns can be expected from companies that are small, value (high book-to-market), and profitable that are not embarking on major growth initiatives. (DO YOU KNOW ANY OF THESE???)

FF 5-Factor Model Definition:

$$R_{it} - R_{Ft} = a_i + b_i (R_{Mt} - R_{Ft}) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + e_{it}$$

$(R_{Mt} - R_{Ft})$ “Equity Premium” – return of value-weighted **market portfolio** over risk-free rate

SMB_t “Small-Minus-Big” - difference in returns in period t of diversified portfolios of stocks with **low and high market capitalization**

HML_t “High-Minus-Low” - difference in returns in period t of diversified portfolios of stocks with **high and low price to book valuation**

RMW_t “Robust-Minus-Weak” - difference in returns in period t of diversified portfolios of stocks with **robust and weak profitability**.

CMA_t “Conservative-Minus-Aggressive” - difference in returns in period t of diversified portfolios of **conservative and aggressive firms with respect to investment behavior**.

Test: Parameters b_i , s_i , h_i , r_i , c_i are “true” values rather than estimates, so if factor exposure capture all cross-sectional variation in expected returns, the intercept a_i is zero for all securities and portfolios i

FF 5-Factor Model Performance:

Average monthly return over one-month Treasury bill of **double-sorted value-weighted portfolios** (1963-2012). **Size**: all NYSE+NASD+AMEX stocks, breakpoints by market capitalization using NYSE stocks. **Profitability**: revenues minus cost of goods sold, minus selling, general, and administrative expenses, minus interest expense all divided by book equity. **Investment**: change in total assets from the fiscal year ending in year t-2 to the fiscal year ending in t-1, divided by t-1 total assets (change in book provides similar results).

Size & Book	Low	2	3	4	High
Small	0.19	0.76	0.8	0.97	1.11
2	0.42	0.68	0.9	0.9	0.97
3	0.45	0.73	0.75	0.85	1.03
4	0.56	0.53	0.68	0.81	0.81
Big	0.42	0.48	0.44	0.52	0.58

Value effect is stronger among small caps

Size & Profitability	Low	2	3	4	High
Small	0.51	0.89	0.86	0.89	0.82
2	0.54	0.74	0.79	0.77	0.93
3	0.49	0.73	0.67	0.73	0.89
4	0.51	0.61	0.58	0.66	0.78
Big	0.33	0.28	0.39	0.43	0.54

Size effect mostly shows in low returns of megacaps.

Q2-4 in profitability have similar returns

Size & Investment	Low	2	3	4	High
Small	0.96	0.94	0.92	0.86	0.29
2	0.88	0.86	0.89	0.85	0.43
3	0.86	0.88	0.76	0.79	0.45
4	0.74	0.68	0.68	0.71	0.51
Big	0.67	0.49	0.45	0.44	0.37

Except megacaps, only High-Investment firms show drastically lower returns

FF 5-Factor Model Performance:

How to deal with correlations of B/M, Profitability and Investment?

❑ High B/M tend to have low profitability and low investment

❑ Large low B/M (growth stocks) tend to be profitable and invest aggressively....

Double sort do not isolate value, profitability and investment effects in average returns....

So, **triple** sort. 2 size groups (Small & Big) using NYSE median and 4 groups for other 2 variables (32 portfolios)

❑ Correlation – uneven allocation of names in 32 portfolios...

❑ “Solution”: different breakpoints for profitability and investment for Small & Big stocks.

Also, can do 4-way sorting. (3^4) = 81 poorly-diversified portfolios, so doing (2^4) = 16

Value and profitability (HML & RMW) remains 63% correlated:

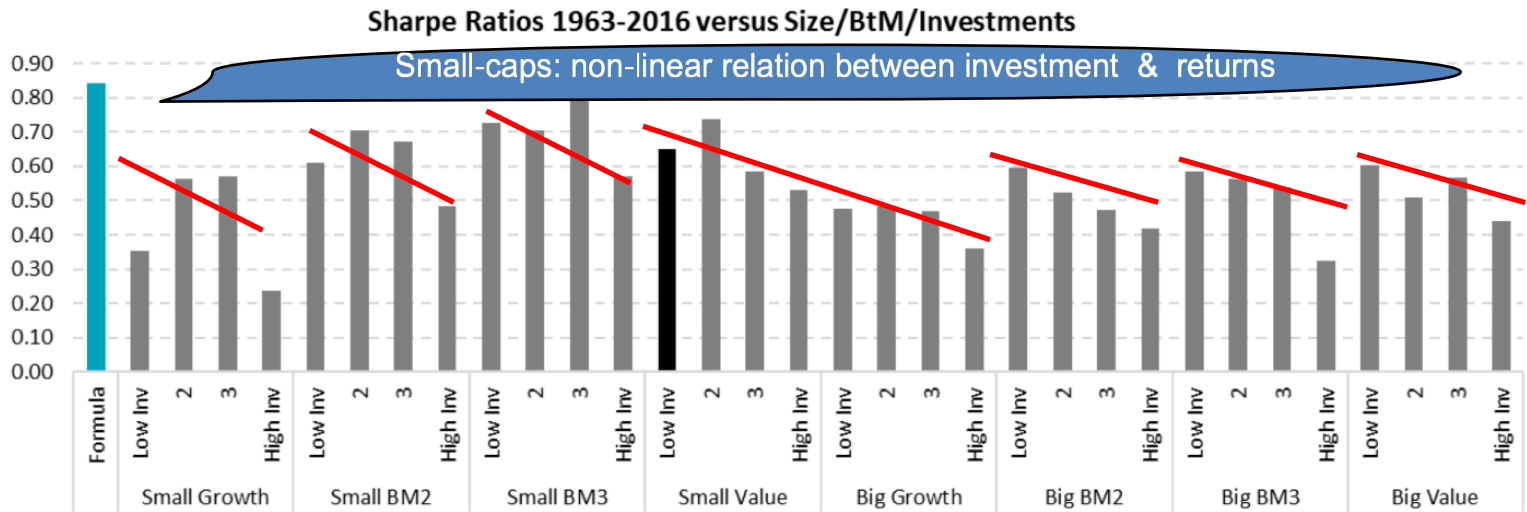
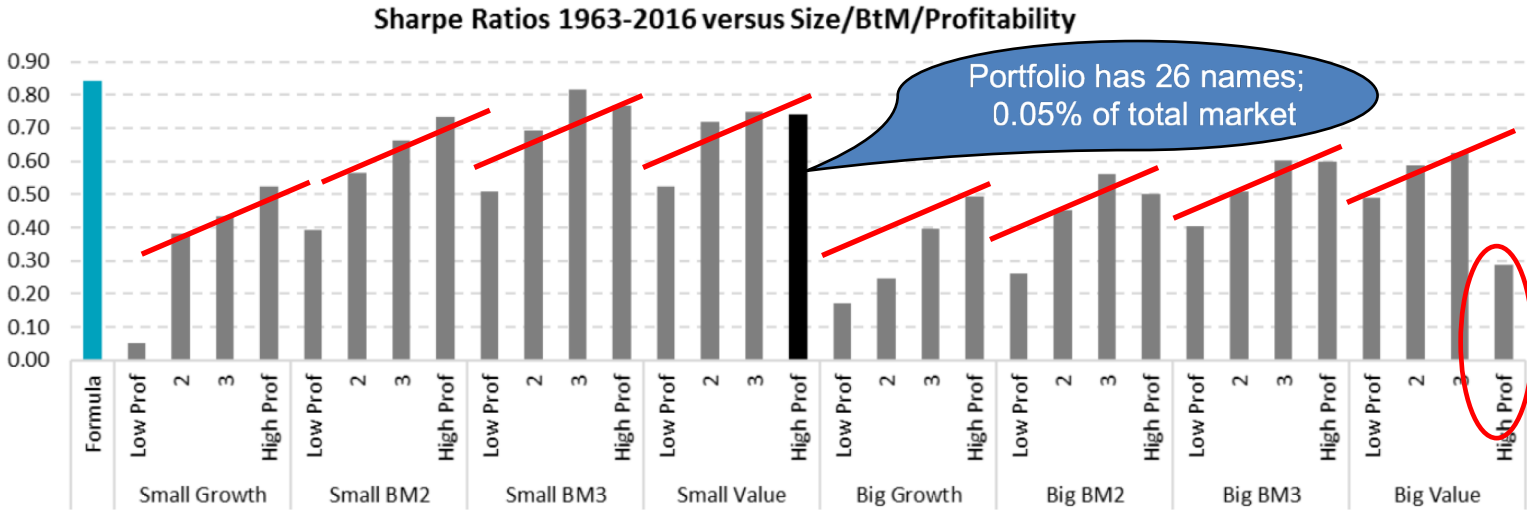
❑ Small non-profitable growth stocks are numerous and held short in both HML and RMW.

	2x2x2x2 factors				
	$R_M - R_F$	<i>SMB</i>	<i>HML</i>	<i>RMW</i>	<i>CMA</i>
Mean	0.46	0.29	0.30	0.26	0.15
Std Dev	4.51	2.93	2.18	1.53	1.18
<i>t</i> -statistic	2.47	2.44	3.39	4.10	3.08

	SMB	HML	RMW	CMA
MKT	25%	-34%	-27%	-43%
SMB		-21%	-33%	-21%
HML			63%	37%
RMW				15%



The Conservative Formula:
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3145152



GRS test is usually rejected. Can be used for relative performance of the models, e.g. pick the model with lowest GRS statistics. For 5 factor portfolios intercept is lower than for 3-factor portfolios.

<https://fnce.wharton.upenn.edu/wp-content/uploads/2017/10/BarillasShanken-9.21.17.pdf>

We have derived a Bayesian asset-pricing test that requires a prior judgment about the magnitude of plausible model deviations or "alphas" and is easily calculated from the GRS F-statistic. Given a set of candidate traded factors, we develop a related test that permits an analysis of Bayesian model comparison, i.e., the computation of model probabilities for the collection of all possible pricing models that are based on subsets of the given factors. This ability to simultaneously compare many models is an attractive aspect of the Bayesian approach.

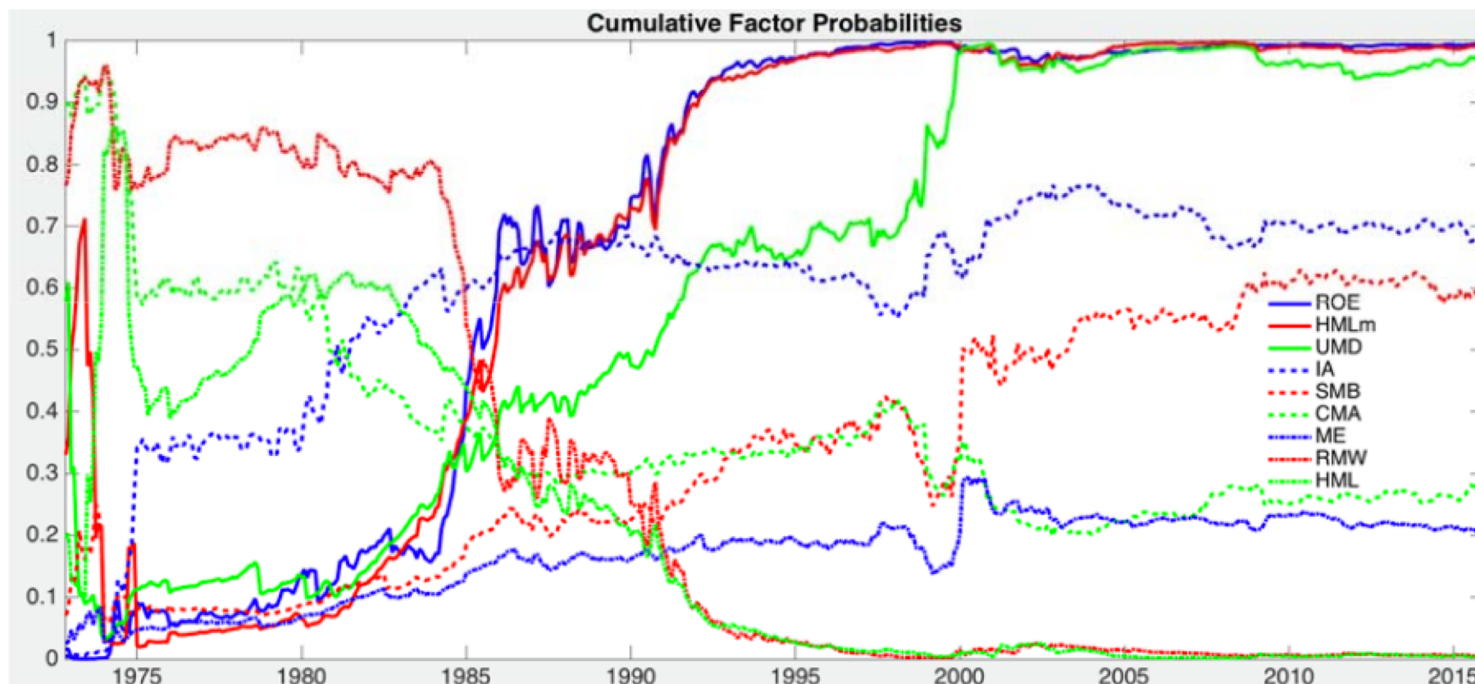
Intercepts for each Factor in the Six-Factor Model on the other Factors

The table presents annualized alphas from regressions of each factor on the other five factors in the model {Mkt SMB ROE IA UMD HML^m}. This model has the highest posterior probability in the 10-factor analysis. The sample period is from January 1972 to December 2013. T-statistics are reported in parentheses.

Factor	SMB	ROE	IA	UMD	HML ^m
Alpha (t-statistic)	4.70 (3.01)	6.51 (5.90)	1.17 (1.51)	6.44 (4.05)	5.59 (5.05)

Comparing Asset Pricing Models (F. Barillas and J. Shanken)

<http://faculty.chicagobooth.edu/eugene.fama/teaching/Reading%20List%20and%20Notes/GRS.pdf>



ROE - profitability (Income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock – more timely – (FF earnings for the fiscal year ending in calendar year $t-1$, divided by the corresponding $t-1$ book equity).

HMLm - AQR's version of HML (more timely update, e.g. most recent monthly stock price in denominator)

UMD – Carhart's 12-1 momentum

IA - [Kewei Hou](#), [Chen Xue](#), [Lu Zhang](#) investments-to-assets (annual change in total assets (Compustat annual item AT) divided by one-year lagged assets (FF – same change from $T-2$ to $T-1$ divided by most recent assets)

ME – another version of size (differs in precise sorts used to construct “Big” and “Small”)

Alternative Factor Definitions

Carhart, Mark, 1997, On Persistence in Mutual Fund Performance, Journal of Finance 52, 57–82

<http://personal.psu.edu/qxc2/fin597/5-carhart-JF.pdf>

Digesting Anomalies: An Investment Approach [Kewei Hou](#), [Chen Xue](#), [Lu Zhang](#)

<http://www.nber.org/papers/w18435>

AQR's more "recent" value factor:

<https://www.aqr.com/Insights/Research/Journal-Article/The-Devil-in-HMLs-Details>

A Comparison of New Factor Models: [Kewei Hou](#), [Chen Xue](#), [Lu Zhang](#)

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2520929

Using the hundreds of significant anomalies in the broad cross section, we compare the performance of a large array of empirical asset pricing models, including the CAPM, the Fama-French (1993) three-factor model, the Carhart (1997) four-factor model, the Pastor-Stambaugh (2003) four-factor model, the Hou-Xue-Zhang (2015) q-factor model, and the Fama-French (2015) five-factor model. The q-factor model and the five-factor model are the two best performing models.

Out-of-Sample Model Performance

The table presents in- and out-of-sample results on the performance of the three top-ranked models by our Bayesian procedure with prior Sharpe multiple 1.5, as well as four prominent models: CAPM, FF3, FF5 and HXZ. The first column gives the in-sample Sharpe ratio over the whole sample 1972-2015, denoted T. Columns labeled EST report the in-sample Sharpe ratio for the two estimation periods which correspond to half (T/2) and two-thirds (2T/3) of the sample. Columns labeled PERF and PERFw report the out-of-sample Sharpe ratios. PERF uses the out-of-sample period to compute the tangency-portfolio weights while PERFw uses the estimation-period returns to determine the tangency-portfolio weights and applies those weights to the out-of-sample returns. The three best models always include Mkt, ROE, HML^m and UMD. Those based on the whole sample also include, respectively: (SMB IA), (SMB CMA) and (ME IA). Those based on half of the sample (T/2) include: (IA), (SMB IA) and CMA, and the first two-thirds of the sample (2T/3): (SMB IA), IA and (ME IA).

Model	T	T/2			2T/3		
	Sample SR	EST	PERF	PERFw	EST	PERF	PERFw
1	0.488	0.778	0.345	0.235	0.596	0.312	0.276
2	0.486	0.815	0.370	0.248	0.567	0.284	0.256
3	0.483	0.779	0.350	0.237	0.592	0.298	0.269
CAPM	0.134	0.094	0.138	0.138	0.118	0.109	0.109
FF3	0.169	0.277	0.165	0.129	0.257	0.147	0.122
FF5	0.356	0.487	0.372	0.202	0.354	0.367	0.263
HXZ	0.373	0.613	0.359	0.250	0.489	0.288	0.271

Too Many or Too Few Market Anomalies? – Replicating Anomalies

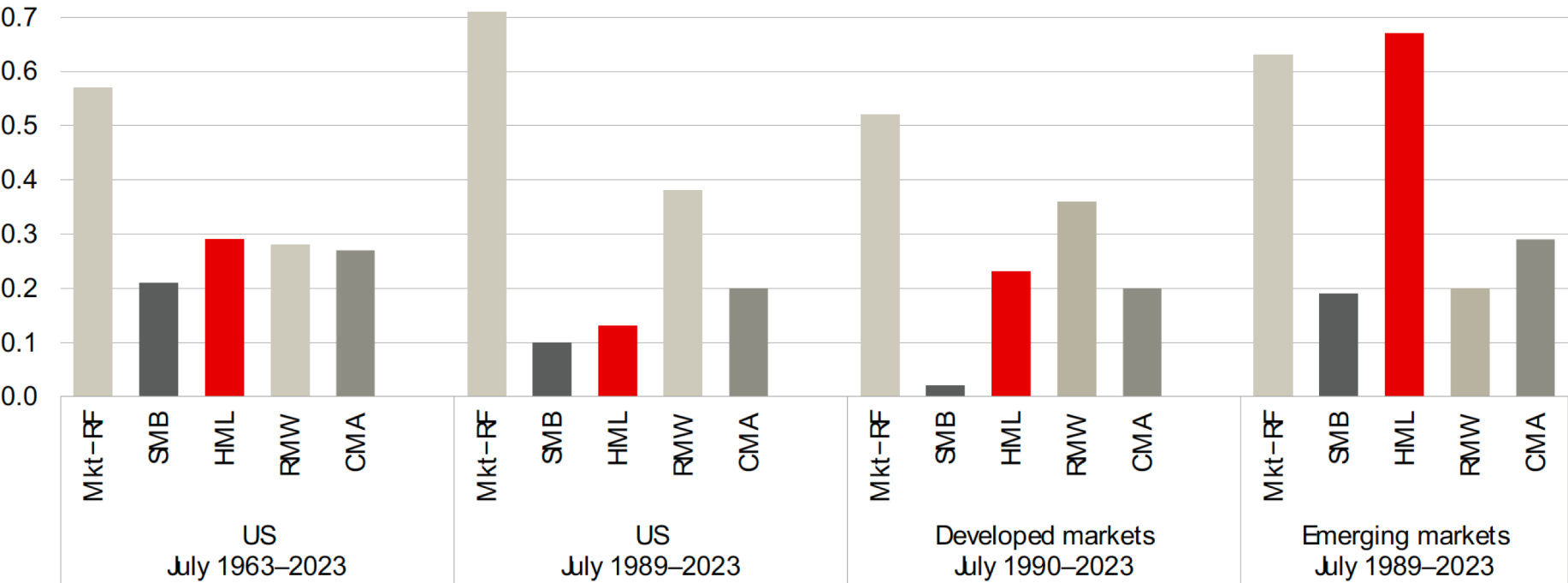
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2961979 (K. Hou, C Xue, L. Zhang, 2017)

The anomalies literature is infested with widespread p-hacking. We replicate this literature by compiling a large data library with 447 anomalies. With microcaps alleviated via NYSE breakpoints and value-weighted returns, 286 anomalies (64%) including 95 out of 102 liquidity variables (93%) are insignificant at the 5% level. Imposing the t-cutoff of three raises the number of insignificance to 380 (85%). Even for the 161 significant anomalies, their magnitudes are often much lower than originally reported. Among the 161, the q-factor model leaves 115 alphas insignificant (150 with $t < 3$). In all, capital markets are more efficient than previously recognized.

Arb1	Abnormal returns around earnings announcements
Ilr1	Industry lead-lag effect
dWc	Change in net noncash working capital
Oa	Operating accruals
Dac	Discretionary accruals
Cop	Cash-based operating profits-to-assets
Cla1	Cash-based operating profits-to-lagged assets (1 month holding period)
Claq12	Cash-based operating profits-to-lagged assets (12 month holding period)
Ra[25]	Heston & Sadka (2008) calendar momentum: (years 2-5 lagged returns)
Ra[610]	Heston & Sadka (2008) calendar momentum: (years 6-10 lagged returns)
Ra[1115]	Heston & Sadka (2008) calendar momentum: (years 11-15 lagged returns)
Ra[1620]	Heston & Sadka (2008) calendar momentum: (years 16-20 lagged returns)
Isq1	Idiosyncratic skewness of the residual returns per q-factor model (1 month holding period)

Style factors around the world. 100 years

Average return difference per month (%)



Key: Mkt – RF is the Market factor (Market return minus Risk Free interest rate); SMB is the size factor (Small Minus Big); HML is the value factor (High book-to-market Minus Low book-to-market); RMW is the profitability factor (Robust firms Minus Weak firms); CMA is the investment factor (Conservative investment Minus Aggressive investment).

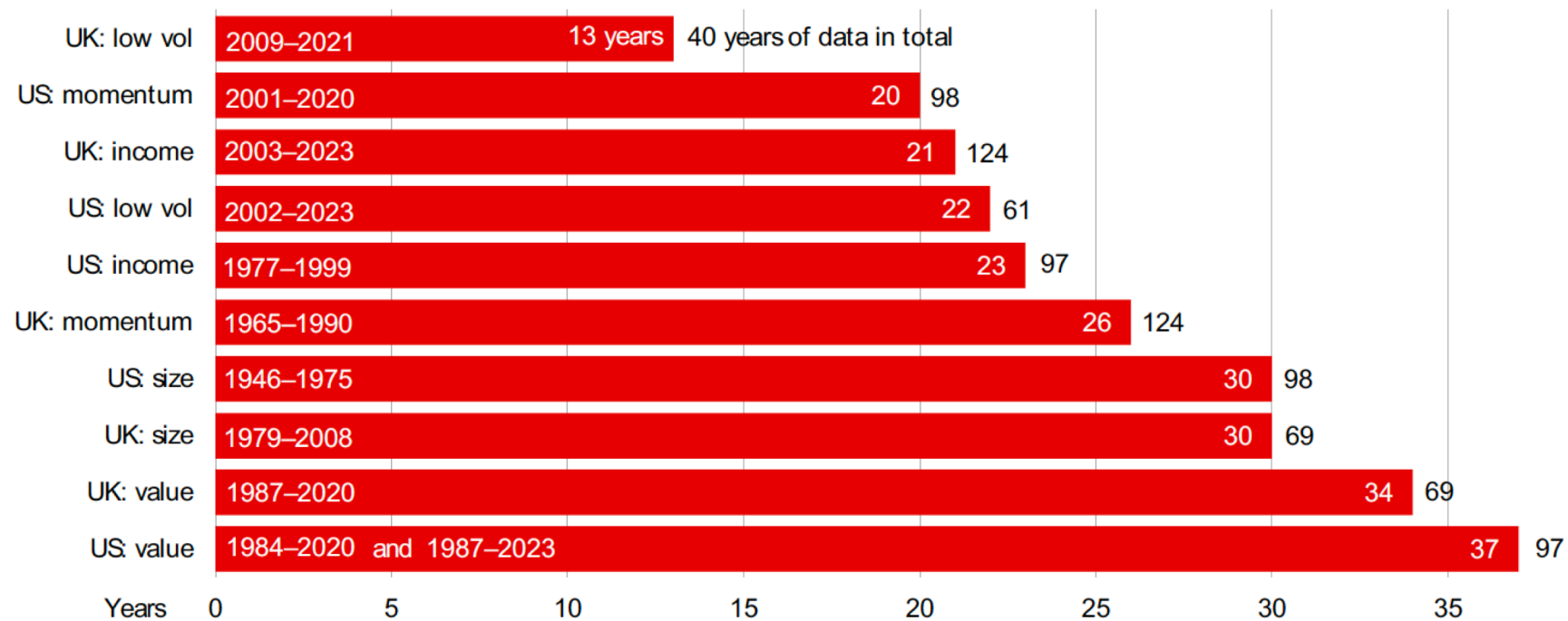
Style factors around the world. 100 years

Factor	Historical premium (%) (number of years)		
	United States	United Kingdom	All countries
Size: Small-cap versus large-cap	1.9 (98)	2.8 (69)	2.5 (45)
Value: High book-to-market versus low book-to-market	2.6 (97)	4.5 (69)	1.9 (40)
Income: High yield versus low yield	1.4 (97)	2.4 (124)	3.0 (52)
Momentum: Winners minus losers	7.6 (97)	9.9 (124)	8.2 (40)
Low volatility: Low volatility versus high volatility	4.4 (61)	7.3 (40)	n/a

Style factors around the world. 100 years

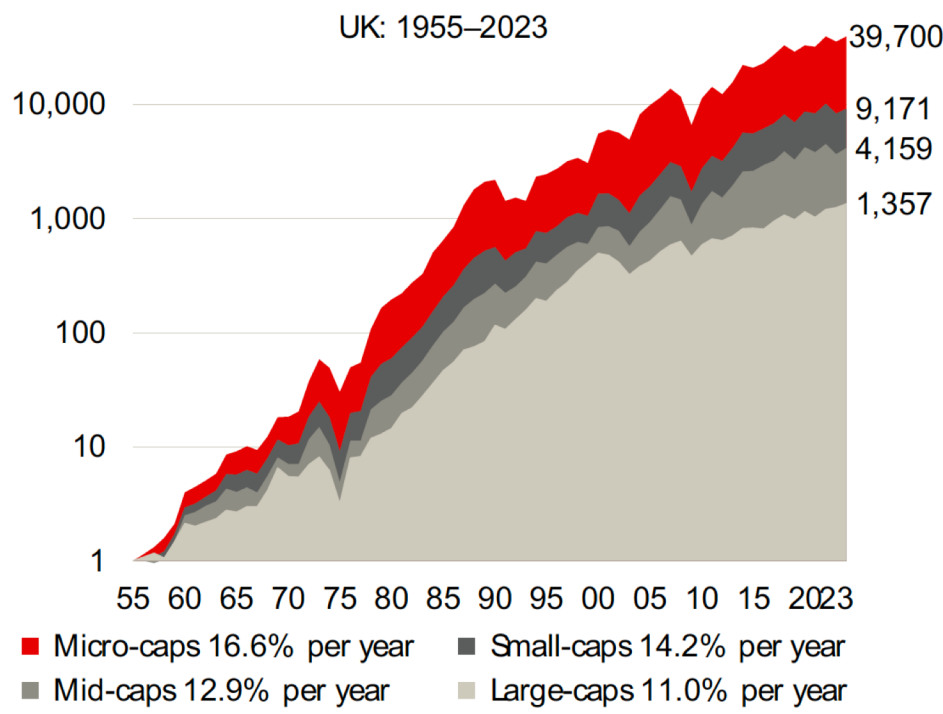
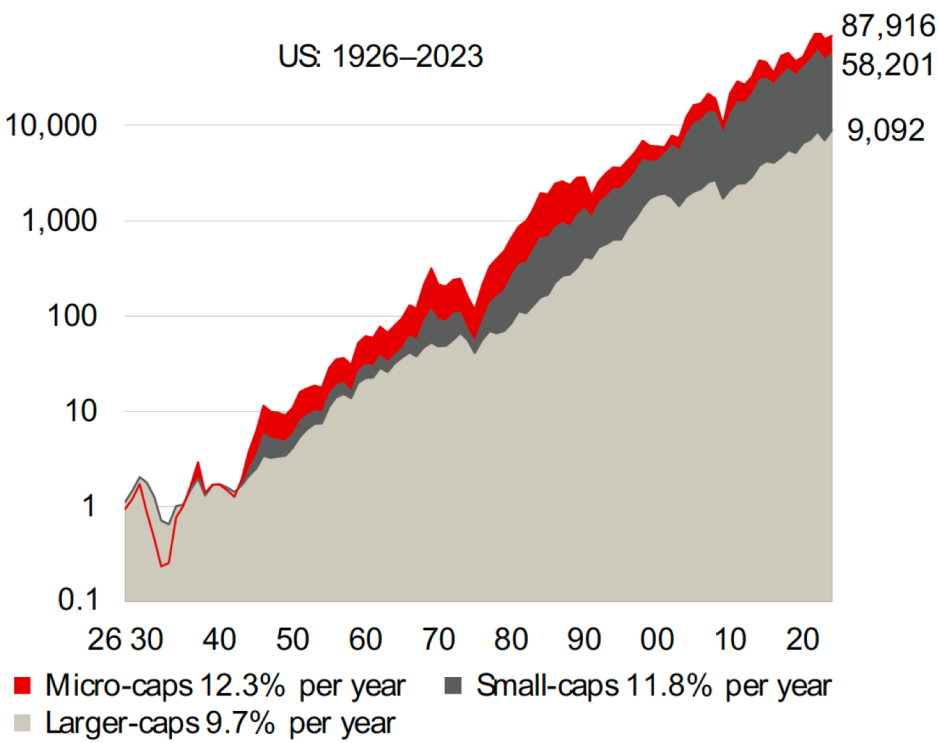
	1900s	1910s	1920s	1930s	1940s	1950s	1960s	1970s	1980s	1990s	2000s	2010–23	All years
US Highest			Mom 20.6	Size 3.9	Value 9.7	Mom 8.6	Mom 10.6	Value 9.6	Low vol 16.6	Mom 10.2	Low vol 11.3	Mom 7.2	Mom 7.6
			Value -2.3	Mom 1.0	Mom 7.2	Value 3.7	Value 3.9	Mom 8.0	Mom 4.9	Low vol 0.6	Size 10.8	Size 0.0	Low vol 4.4
			Income -7.0	Income -2.1	Income 5.7	Income 0.4	Size 3.2	Size 5.3	Value 4.9	Value -2.6	Value 6.9	Income -0.8	Value 2.6
			Size -12.4	Value -4.7	Size 5.1	Size -0.1	Income 1.6	Low Vol 5.4	Income 4.5	Size -3.5	Income 5.8	Low vol -1.6	Size 1.9
Lowest							Low vol -5.1	Income 4.9	Size 0.2	Income -4.0	Mom -1.9	Value -3.1	Income 1.4
UK													
Highest	Mom 14.1	Mom 22.5	Mom 17.2	Mom 13.5	Income 4.4	Mom 19.9	Value 9.5	Value 10.5	Value 6.2	Mom 9.3	Low vol 18.7	Mom 6.8	Mom 9.9
	Income 3.4	Income 2.2	Income 5.7	Income -0.6	Mom 0.6	Value 9.9	Mom 6.6	Size 8.1	Income 4.1	Low vol 3.3	Mom 11.9	Low vol 3.2	Low vol 7.3
						Size 6.7	Size 3.2	Income 5.5	Mom 3.7	Value -0.9	Income 6.1	Size 2.7	Value 4.5
						Income 1.7	Income 1.3	Mom -3.2	Low vol 3.5	Income -1.8	Size 3.5	Income -0.9	Size 2.8
Lowest									Size 1.6	Size -3.7	Value 3.4	Value -1.0	Income 2.4

Style factors around the world. 100 years



Style factors around the world. 100 years

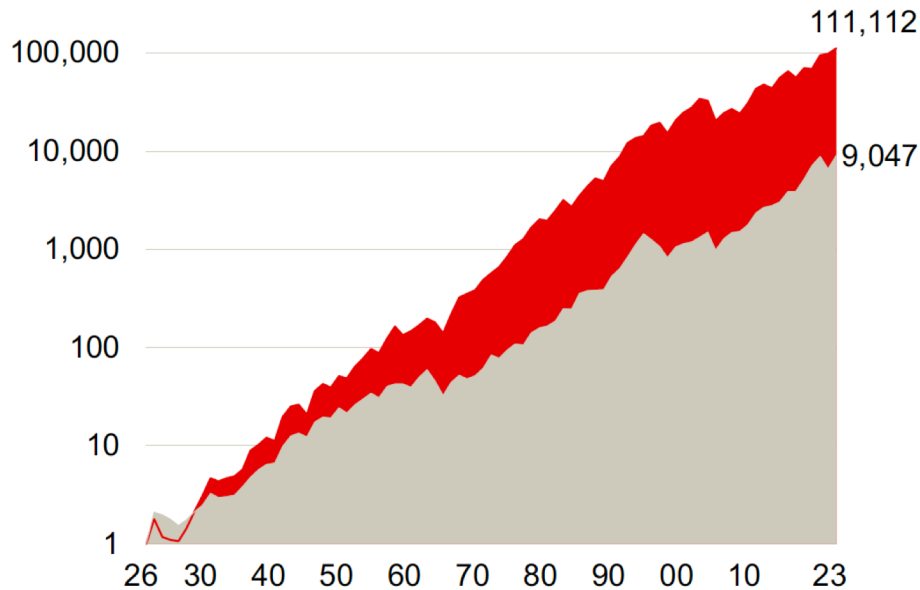
Size



Style factors around the world. 100 years

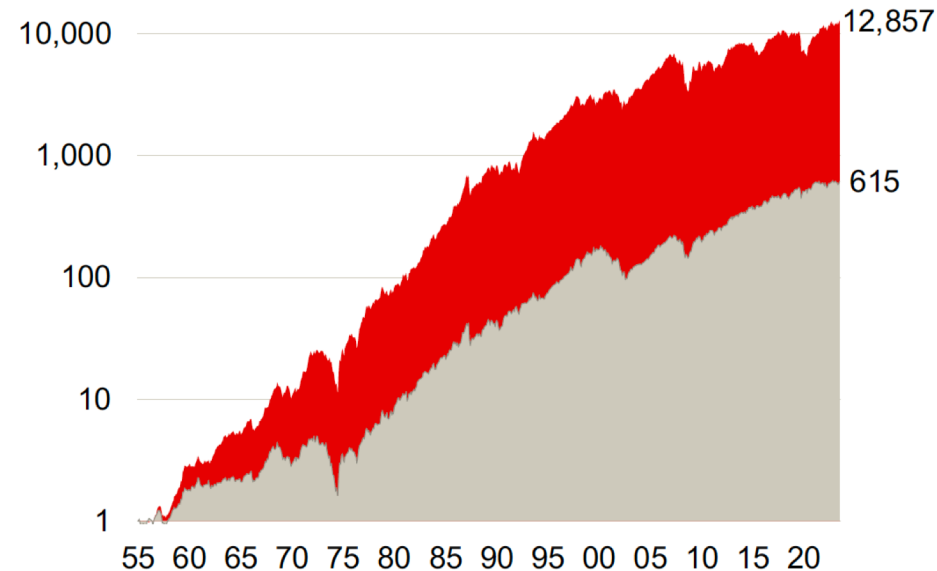
Value

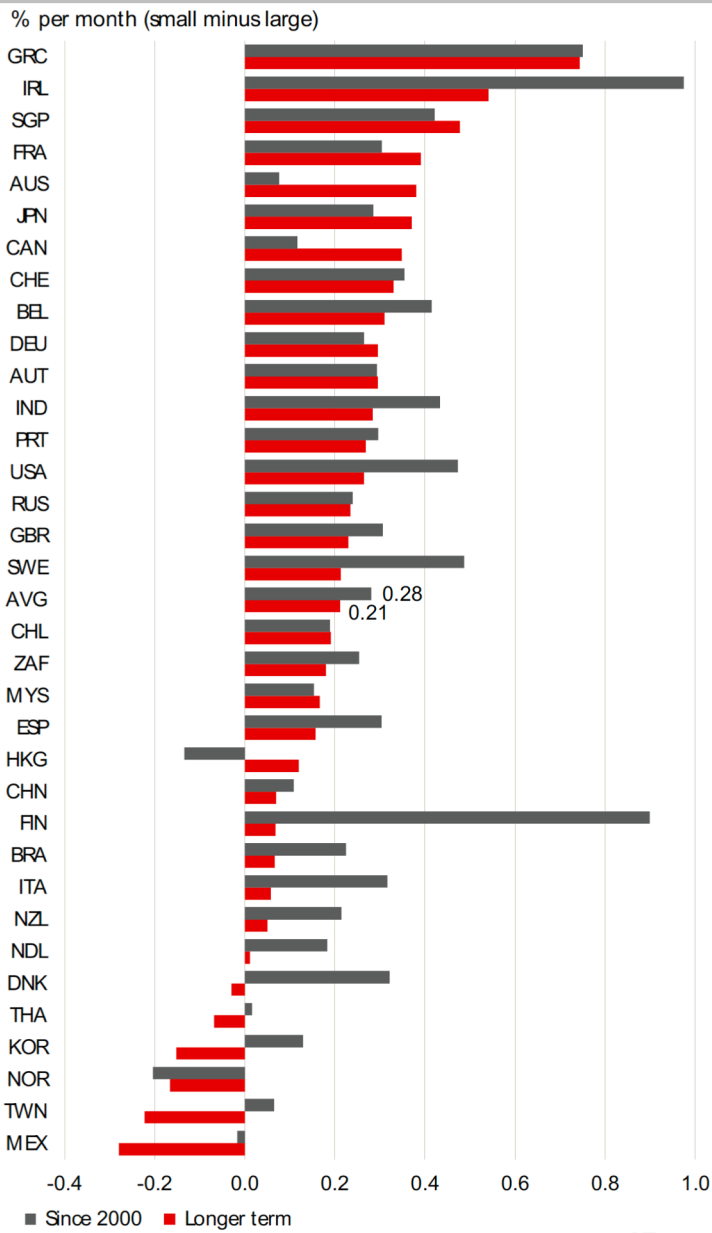
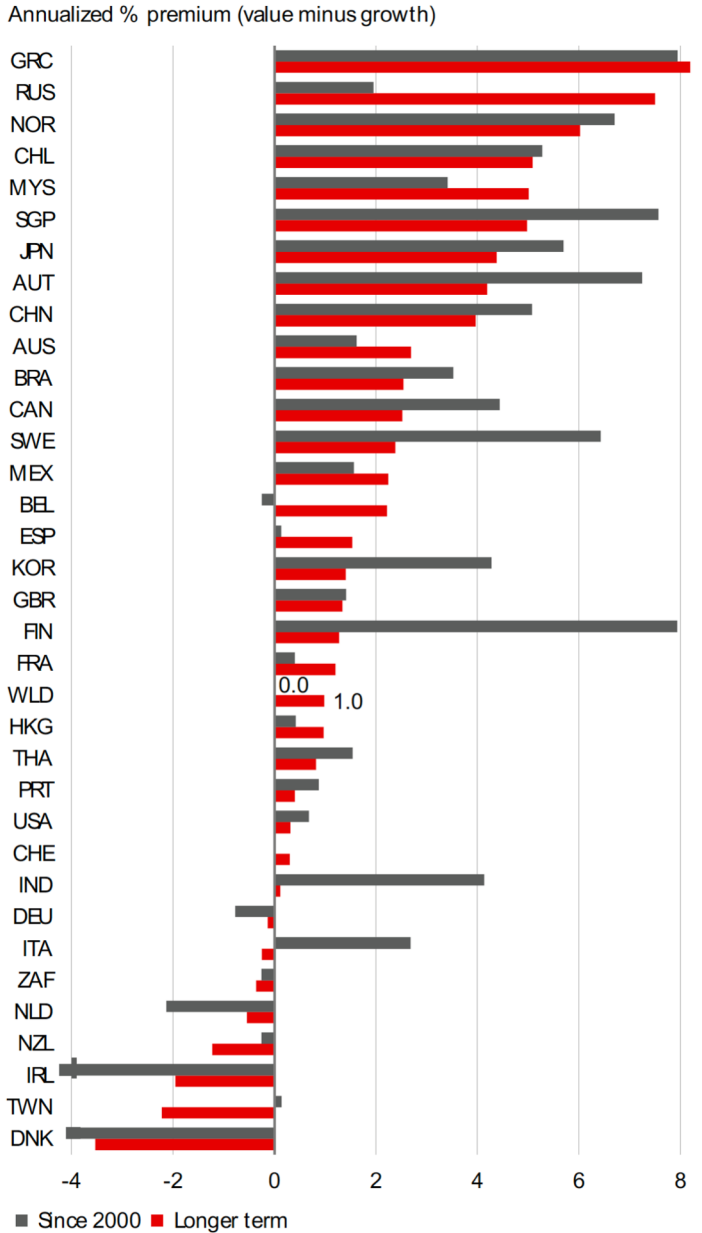
US: 1927–2023



■ High book-to-market 12.7% p.a. ■ Low book-to-market 9.8% p.a. ■ High book-to-market 14.8% p.a. ■ Low book-to-market 9.8% p.a.

UK: July 1955–December 2023

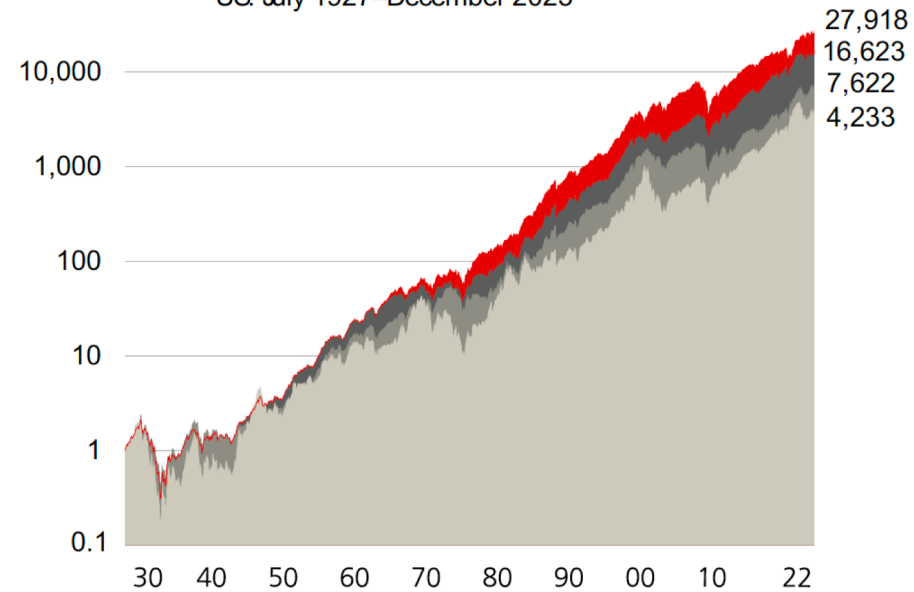




Style factors around the world. 100 years

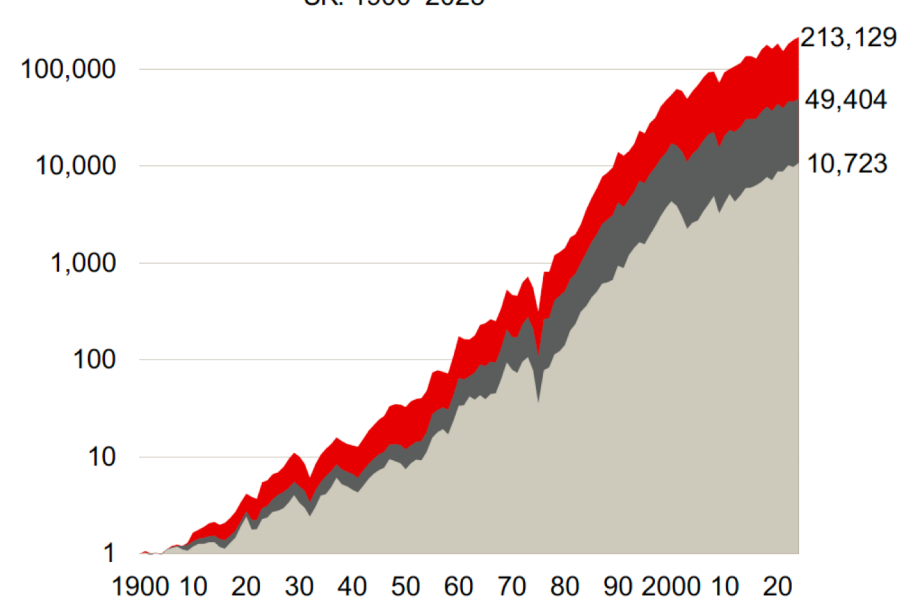
Dividend yield

US: July 1927–December 2023



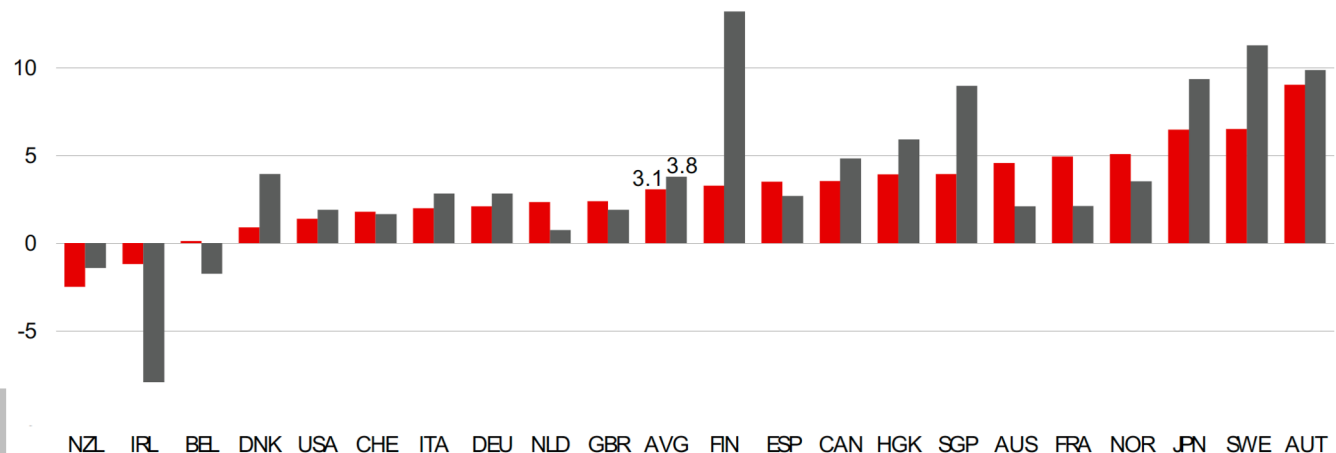
■ High yield 11.2% p.a. ■ Medium yield 10.6% p.a.
■ Low yield 9.7% p.a. ■ Zero yield 9.0% p.a.

UK: 1900–2023



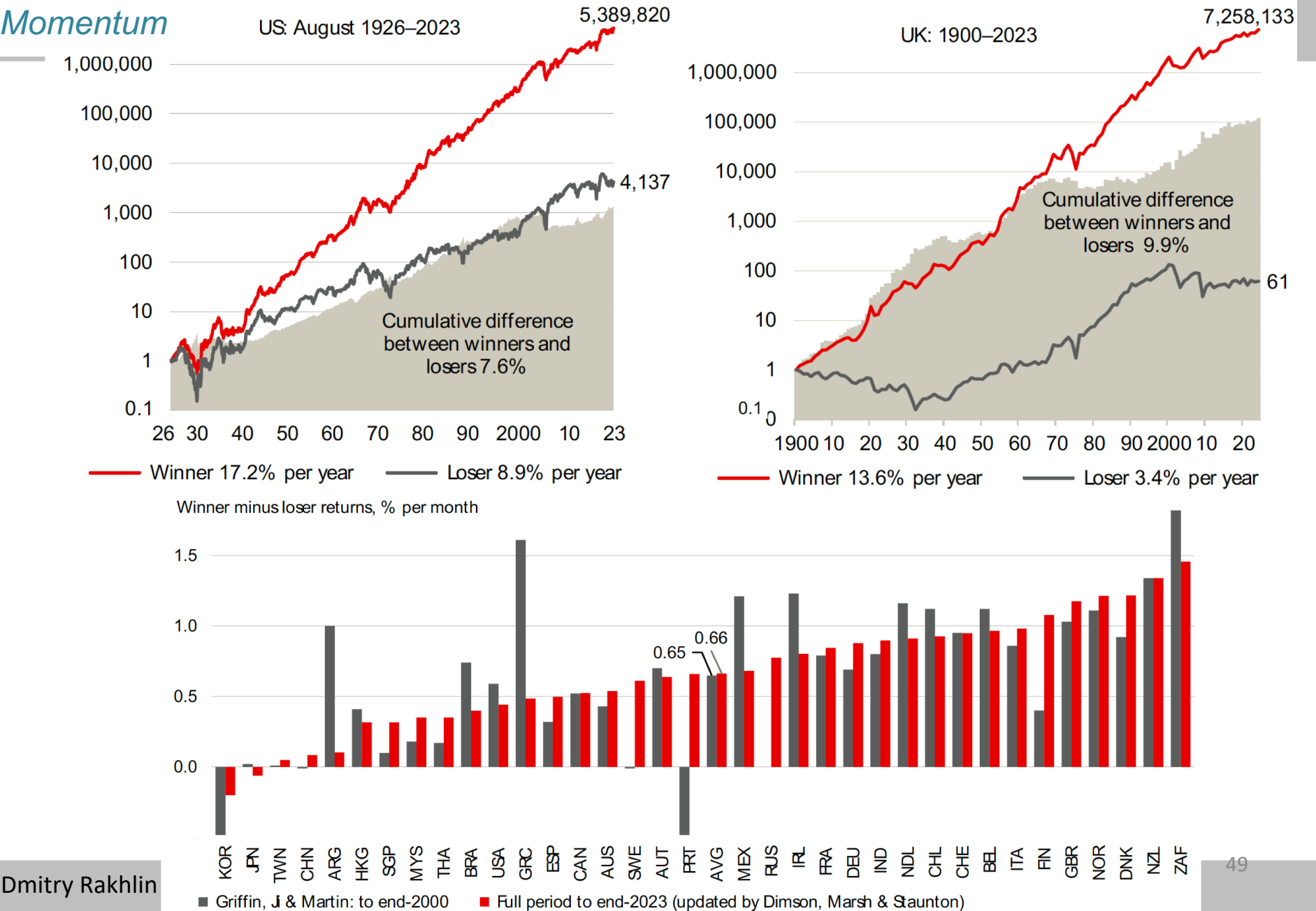
■ High yield 10.4% p.a. ■ UK market 9.1% p.a. ■ Low yield 7.8% p.a.

Yield premium (%) (high yield minus low yield)



Style factors around the world. 100 years

Baruch



Style factors around the world. 100 years

Low Risk (a.k.a. – "bet against beta")

